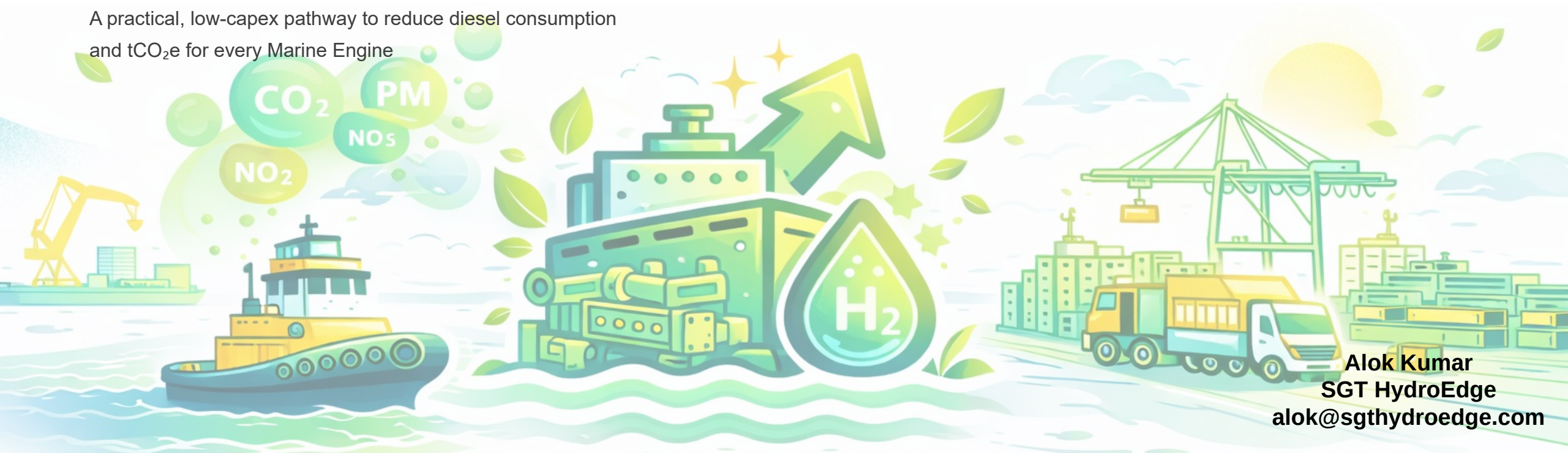


REDUCING GHG EMISSIONS FROM MARINE ENGINES USING GreenMarine™ HYDROGEN FUEL ASSIST RETROFIT

DECARBONISATION WITH PROFIT

A practical, low-capex pathway to reduce diesel consumption and tCO₂e for every Marine Engine



Marine Industry at a Turning Point



- **IMO's Ambitious 2050 Net-Zero Target**

The IMO has established a goal to achieve net-zero greenhouse gas emissions from international shipping by 2050, requiring immediate action.



- **Regulatory Compliance Pressure**

Stricter emission limits under MARPOL Annex VI and impending carbon pricing mechanisms are transforming operational requirements.



- **Legacy Fleet Challenge**

90% of vessels operate with legacy diesel engines that cannot be easily replaced, creating an urgent need for retrofit solutions.



- **Financial Imperative**

Fleet operators face mounting pressure to maintain profitability while investing in sustainability technologies.



- **Beyond CO₂ Concerns**

Regulations increasingly target multiple pollutants including NO_x, SO_x, and particulate matter, requiring comprehensive solutions.

Industry Status Quo

- **90% of Global Fleet:** Operates with legacy engines
- **Immediate Need:** Retrofit Solutions for Decarbonization
- **Key Challenge:** Balancing Profitability with Sustainability

The Marine Sustainability Challenge



Regulatory Pressure

Stricter emission limits under MARPOL Annex VI, impending carbon pricing mechanisms, and complex global compliance requirements.



The Compliance Dilemma

Operators must balance immediate regulatory needs with long-term sustainability goals while maintaining commercial viability.



Commercial Pressure

Growing customer demand for green shipping, increased ESG scrutiny from investors, and brand reputation risks in sustainability performance.



Traditional Solutions Drawbacks

High capital expenditure, significant vessel modifications, and extended implementation timelines limit adoption of conventional approaches.



Operational Pressure

Volatile fuel costs directly impact profitability, aging fleets require retrofitting solutions, and limited options exist for immediate implementation.

Introducing SGT -GreenMarine™



Revolutionary Hydrogen-Assisted Combustion

Precise hydrogen injection optimizes fuel burning for maximum efficiency and minimal emissions.



Immediate Performance Benefits

3-15% fuel savings, 70%+ particulate matter reduction, and significant NOx decrease across all engine loads.



Simple Retrofit Solution

Non-invasive installation requires no engine modifications, preserving warranty and classification.



Versatile Application

Compatible with auxiliary engines and high-speed propulsion systems across various vessel types.



Real-Time Performance Monitoring

IoT-enabled system provides continuous data on fuel savings and emissions reduction.



Extended Engine Life

Lower operating temperatures and reduced carbon deposits extend maintenance intervals.



System Benefits

- 3-15% reduction in diesel consumption based on load
- Significant reduction in PM, NOx, CO, and HC emissions
- Optimizes combustion without engine modification
- No need for high-pressure hydrogen storage-Inherently safe

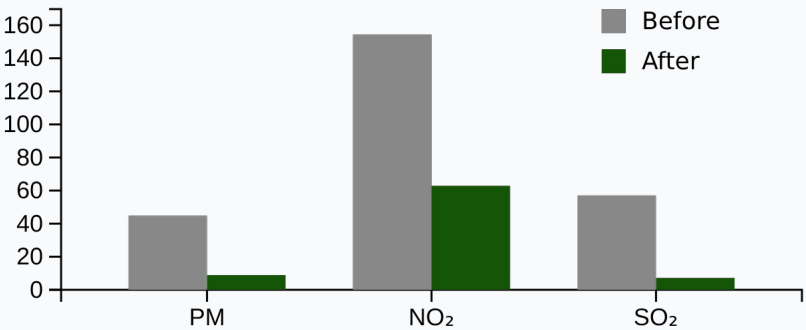
Proven Validation for Large DG Sets (Aux. Engines)

Two Sample Test Results

Independent NABL Testing

⚡ 1500 kVA DG Results

Tested by NABL Accredited Lab



PM Reduction:
80%

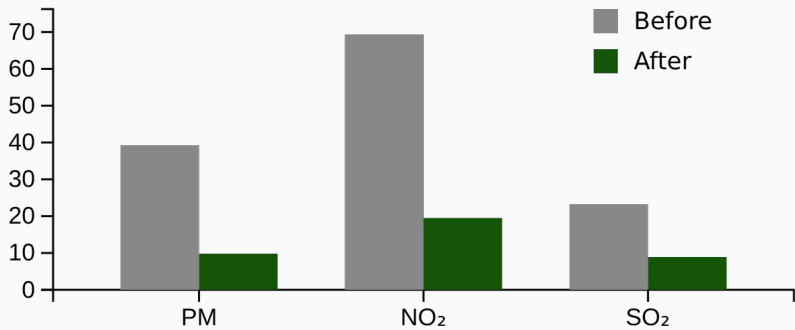
NO₂ Reduction:
59%

SO₂ Reduction:
87%

Fuel Efficiency:
8.5%

⚡ 380 kVA DG Results

Tested by NABL Accredited Lab



PM Reduction:
75%

NO₂ Reduction:
72%

SO₂ Reduction:
62%

Fuel Saving:
3-12% (based on load)

✓ **Combined Results: Significant Emission Reductions & Fuel Efficiency Gains**

8.5% Average Fuel Savings



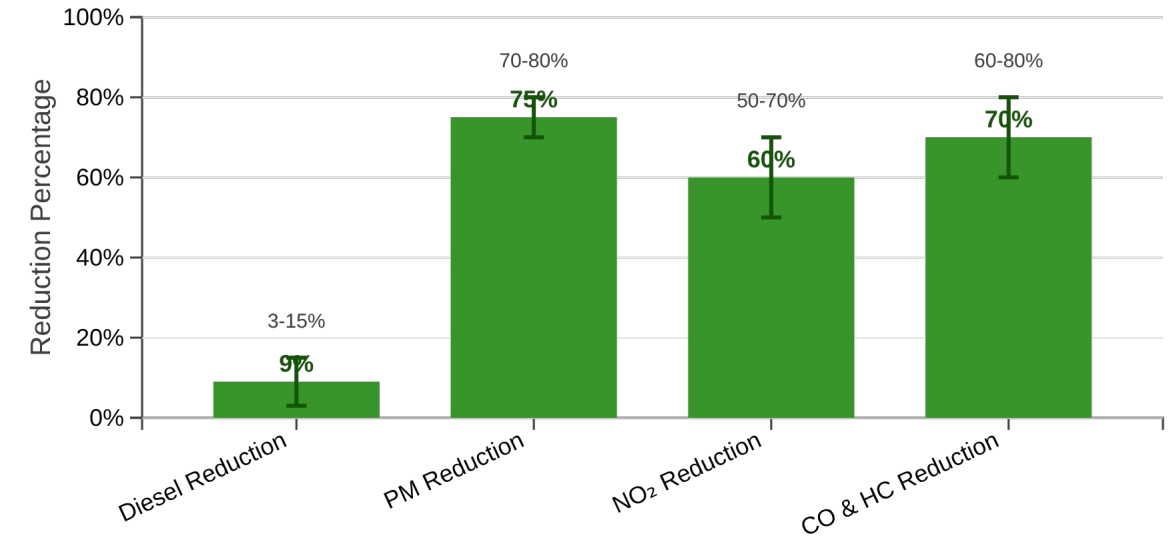
Proven Validation Across DG Sizes

Comprehensive Testing & Validation

-  **Range:** Tested on diesel generators from **125 kVA to 1500 kVA**
-  **Fuel Reduction:** Consistent **3% to 15%** reduction in diesel consumption
-  **Particulate Matter:** **70% to 80%** reduction in PM emissions
-  **NO₂ & Other Emissions:** **50% to 70%** reduction in NO₂ and **60% to 80%** reduction in CO & HC

Predictable Performance

GreenMarine™ demonstrates consistent and reliable performance across all DG sizes, making it a versatile solution for various decarbonization needs.



 **TRL 8-9** technology readiness level

 **Rapid installation** in less than 4 hours

 Compatible with **20-4000 kVA** DG sets

GreenMarine™ Performance on Marine Engines



Auxiliary Engine Applications

Immediate fuel savings of 3-15% for continuous hotel loads, significant port emission reductions, and enhanced engine reliability through cleaner combustion.



High-Speed Propulsion Applications

Optimized performance for vessels with 750-1800 RPM engines including workboats, ferries and offshore support vessels across variable duty cycles.



Operational Benefits

Improved throttle response, smoother engine operation, and compliance without complex exhaust after-treatment systems adding backpressure.



Retrofit Compatibility

Engineered specifically for marine environments with minimal installation footprint and seamless integration with existing systems.



Future Applications

Research underway for adaptation to slow-speed large engines (70-120 RPM) used in container ships and bulk carriers.

How GreenMarine™ Works



Water Electrolysis

System splits water (H_2O) into hydrogen (H_2) and oxygen (O_2) gas through electrolysis



Gas Infusion

Electronically metered
Generated gas mixture
blended into diesel engine's
air intake manifold



Combustion Optimization

Metered Hydrogen improves
combustion by increasing
flame speed and reducing
ignition delay

Key Benefits

-  **Lower Diesel Consumption** - Optimized combustion leads to more efficient fuel usage & corresponding tCO₂e reduction.
-  **More Complete Burn** - Ensures thorough utilization of diesel fuel
-  **Reduced Emissions** - Decreased PM, NO₂, CO, and HC emissions
-  **Stable Power Generation** - Particularly under varying load conditions

Smart Control & Safety Architecture



Adaptive Hydrogen Dosing

Intelligently adjusts hydrogen dosage based on real-time engine conditions for optimal performance and safety

No Hydrogen Storage

Hydrogen produced on-demand and immediately consumed, eliminating risks associated with high-pressure storage



Built-in Safety Logic

Engineered for continuous remote operation with comprehensive safety protocols for demanding environments



Integrated Monitoring

Seamlessly integrated with SGT GreenVision™ for real-time monitoring of fuel usage, load profiles, and emission signatures



Multiple Shutdown Protections

Includes safeguards against overheating, backflow, and electrical anomalies to ensure system integrity



Proven Technology

SGT's Smart Control Architecture has been successfully tested & deployed across various industrial applications, ensuring reliable operation in offshore, 24/7 environments

Dual-Pathway CO₂ Reduction

Direct Fuel Savings & N₂O Avoidance

Direct Fuel Savings

3-15% fuel efficiency improvement

GreenMarine™ delivers 3-15% fuel efficiency improvement through optimized combustion, directly reducing CO₂ emissions and operational costs.



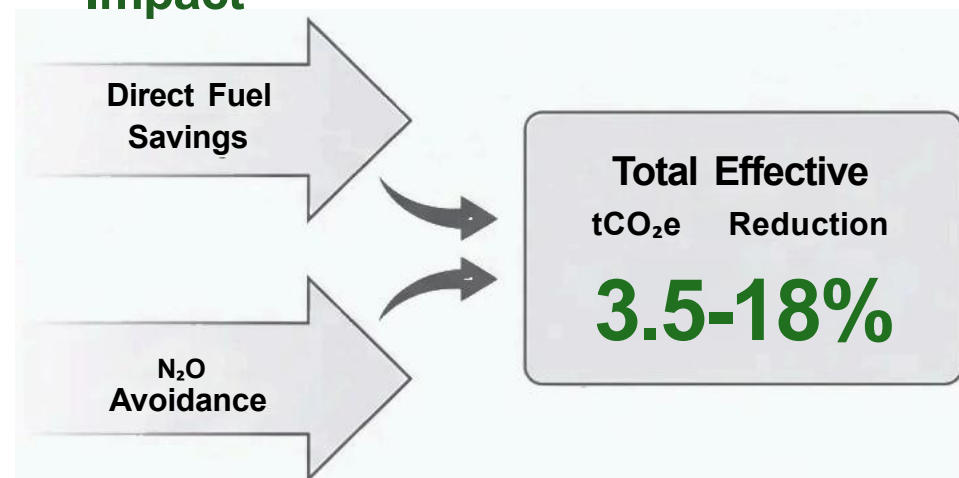
N₂O Emission Avoidance

Avoids SCR N₂O emissions

By eliminating the need for SCR systems that produce N₂O (a potent greenhouse gas), GreenMarine™ prevents additional climate impact.



Combined tCO₂e Impact



No Fuel Switching Required

Achieve meaningful carbon reductions while continuing to use conventional marine fuels, avoiding costly alternative fuel infrastructure.

Measurable Climate Benefits

All emission reductions are quantifiable and reportable for regulatory compliance and ESG reporting requirements.

Stack Emission Data from Real Engines

3-12% fuel efficiency improvement

Avoids N₂O emissions

1500 KVA GENERATOR – STACK EMISSION DATA

Sr No	Measuring Parameter	Units	Testing Standard / Protocol	Limit (mg/Nm3)	Without Hydrogen	With Oxy-Hydrogen	Improvement %
1	PM	mg/Nm3	IS:11255 (Part 1) RA 2019	75	44.8	8.9	80.13
2	SO2	mg/Nm3	IS:11255 (Part 2) RA 2019	NA	31.4	7.8	75.16
3	NO2	mg/Nm3	IS:11255 (Part 7) RA 2017	710	154.3	62.8	59.3
4	CO2	%	IS:13270 RA 2019	NA	5.6	3.2	42.86
5	CO	%	IS:13270 RA 2019	NA	0	0	NA
6	O2	%	IS:13270 RA 2019	NA	12	15.8	-31.67

380 KVA GENERATOR – STACK EMISSION DATA

Sr No	Measuring Parameter	Units	Testing Standard / Protocol	Limit (mg/Nm3)	Without Hydrogen	With Oxy Hydrogen	Improvement %
1	PM	mg/Nm3	IS:11255 (Part 1) RA 2019	75	39.2	9.8	75
2	SO2	mg/Nm3	IS:11255 (Part 2) RA 2019	NA	23.2	8.6	62.93
3	NO2	mg/Nm3	IS:11255 (Part 7) RA 2017	710	69.3	19.5	71.86
4	CO2	%	IS:13270 RA 2019	NA	4.6	3.3	28.26
5	CO	%	IS:13270 RA 2019	NA	0	0	NA
6	O2	%	IS:13270 RA 2019	NA	12.8	14.3	-11.72

Impact on IMO CII & EEXI Compliance



EEXI Compliance Enhancement

GreenMarine reduces specific fuel consumption by 3-15%, improving EEXI ratings without costly engine power limitations, ensuring vessels meet the required energy efficiency standards.



CII Rating Improvement

Direct fuel efficiency gains translate to better annual carbon intensity scores, helping vessels maintain favorable A-C ratings and avoid falling into lower, non-compliant categories.



Avoiding Operational Penalties

Prevents vessels from falling into D-E ratings that require corrective action plans and face potential commercial restrictions and market devaluation.



Cost-Effective Compliance

Achieves regulatory compliance with positive ROI through significant fuel savings, unlike other solutions that increase operational costs or require expensive retrofits.



No Operational Compromise

Maintains full vessel performance and schedule flexibility while improving environmental ratings, allowing operators to meet goals without sacrificing efficiency or capacity.

GreenMarine™ vs. Traditional Solutions

Feature	GreenMarine™	Scrubber (SOx)	SCR (NOx)
CAPEX Model	Zero(DaaS)-Immediate ROI	High-Major Investment	Medium/High-Significant Investment
OPEX Impact	Reduced with Fuel Savings	Increased with Power & Consumables	Increased with Urea & Catalyst
Emissions Reduced	All emissions reduced including CO ₂ , PM(>70%), NOx	SOx, PM	NOx only
Fuel Efficiency	+3-15% Improvement	Reduced (Backpressure)	Reduced (Backpressure)
Installation Footprint	Minimal-Simple Retrofit	Large-Structural Mods	Significant-Space Claim
Financial Model	Decarbonisation-as-a-Service (DaaS gives Positive Cash Flow)	Traditional Purchase-Long Payback	Traditional Purchase-Long Payback

Direct Cost Comparison: GreenMarine™ reduces overall operational costs, unlike scrubbers and SCR which increase expenses through power demands and consumables.

Multiple Emissions Reduction: Simultaneously targets CO₂, PM, and NOx, offering a comprehensive solution compared to single-pollutant technologies.

Installation & Space Requirements: Offers a minimal footprint and simple retrofit process, avoiding major structural modifications.

Financial Model Advantage: Unique DaaS model provides zero upfront CAPEX, guaranteed performance, and immediate positive cash flow.

Operational Impact: Improves fuel efficiency by 3-15%, avoiding the engine backpressure issues associated with traditional exhaust treatment systems.

Decarbonisation-as-a-Service(DaaS)



Zero-CapEx Solution

Eliminate upfront investment through subscription model that preserves capital for core business needs



Immediate Positive Cash Flow

Monthly service fee structured to be less than fuel savings, ensuring day-one profitability



Performance Guarantee

No performance risk-if the system doesn't deliver promised savings,you don't pay



Performance Guarantee

No performance risk-if the system doesn't deliver promised savings,you don't pay



Hassle-Free Operations

SGT manages all maintenance,monitoring and repairs without burdening your crew



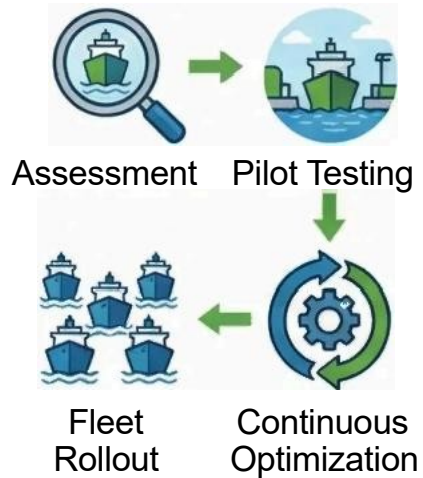
Verified Emissions Reporting

Automated data collection supports ESG goals and regulatory compliance reporting requirements

Addressing Key Concerns

Ensuring a Seamless and Effective Transition

Phased Implementation Approach



Our 4-phase strategy includes assessment, pilot testing, fleet rollout, and continuous optimization.

Zero Disruption Installation



Zero Disruption Installation

Systems are installed during scheduled maintenance periods with minimal operational impact.

Comprehensive Training Support



Comprehensive Training Support

Crew training ensures proper operation and maintenance with 24/7 technical assistance.

Continuous Performance Monitoring



Continuous Performance Monitoring

IoT sensors and cloud analytics provide real-time efficiency data and optimization.

Scalable Fleet-wide Deployment



Scalable Fleet-wide Deployment

Proven methodology allows rapid scaling from single vessel to entire fleet operations.

Contact & Next Steps

Get in Touch

Partner with Saarthi Green to become a decarbonization leader. Our GreenMarine™ Hydrogen Fuel Assist Retrofit offers a practical, high-ROI solution for immediate emission reduction on any marine auxillary engine without requiring significant capital investment or operational disruption.

Take the first step with our 3-60 day Proof of Concept to validate performance on your existing diesel generators and develop a tailored implementation roadmap.



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